

School Bus Tracking System

Developer handover status report — Steps 1–52, Live Tracking, Admin navigation, scheduling, Driver App, and Reports & Analytics.

Audit date: 17 August 2026 · Method: code review plus live API tests on localhost:4001 · Confirmed on TRIP-1005 (bus KDA 123A, driver Daniel Kamau)

52 STEPS REVIEWED	24 DONE / WORKING	21 PARTIAL	7 MISSING
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The core trip loop is working: Recurring Schedule → Trip Instance (Trip ID) → Driver start → GPS → manual check-in → manual check-out → trip complete → Live Tracking while active → history after execution. TRIP-1005 was started, checked in/out, and completed during this audit, so it will not appear as startable in the Driver App for the rest of 17 August 2026.

1. What is done (24 steps)

These items are implemented and were confirmed in code and/or live tests.

STEP	ITEM	EVIDENCE
1	Project foundation & architecture	Monorepo, MongoDB, Express, JWT, Socket.IO, Vite admin/driver/parent apps. API health OK.
2	Authentication & login	Login/logout/JWT tested for school_admin, driver, parent, super_admin, teacher. Invalid credentials return 401.
4	School / organization management	School profile CRUD. Super admin onboards schools; school admin edits name, address, and gate location.
8	Student management	Create, edit, delete, and onboard wizard. 3 students in current data.
9	Parent / guardian management	Parent CRUD. 3/3 students linked to parents.
11	Bus management	Plate, label, seats, status. Bus KDA 123A live-tested.
12	Driver management	Driver users, profiles, contact, bus/route assignment. Daniel Kamau live-tested.
13	Route management	Create/edit/activate routes. Route A — Pipeline live-tested.
15	Student-to-route assignment	3/3 students assigned to a route.
16	Driver-to-route assignment	DriverProfile.assignedRouteIds plus per-schedule driverId.
20	Live GPS tracking foundation	Location pings stored. lat/lng/heading/speed tested on TRIP-1005.
21	Map & vehicle tracking	Admin Live Tracking map with auto-refresh. Active bus showed plate, driver, route, Trip ID.

STEP	ITEM	EVIDENCE
22	Trip management foundation	Trip instances with status, route, bus, driver, and Trip ID (TRIP-nnnn).
23	Driver App (foundation)	Web + APK. Today's instances, start/end, GPS, check-in/out.
24	Trip start	TRIP-1005 started; start GPS stored. Cancelled, completed, and already-active trips cannot start.
25	Trip stop / end	End GPS stored. Complete blocked while a student was still checked in (HTTP 409).
26	Manual student check-in	Pickup stored against TRIP-1005. Live count went to in=1.
27	Manual student check-out	Drop-off stored against TRIP-1005, then trip completed.
29	Parent transport visibility	Parent sees linked child, live trip API, trip history, and notifications.
36	Parent / student notifications foundation	trip_started, kid_picked_up, kid_dropped_off, trip_completed emitted after TRIP-1005.
39	Security & access control	Driver/parent blocked from admin (403). Admin blocked from driver/parent routes.
45	Deployment & production configuration	Render API + Vercel web + env/JWT/Mongo/Mapbox. Remote /health OK.
46	Core system & Driver App handover	Foundation for Steps 47–52 is in place.
49	Student & parent assignment	parentIds on students. Parent visibility confirmed.

2. What is partial (21 steps)

These exist in some form, but they do not fully match the handover document.

STEP	ITEM	WHAT WORKS	WHAT IS STILL INCOMPLETE
3	User & role management	Roles exist: super_admin, school_admin, driver, parent, teacher.	No Users & Roles screen or permission matrix. /admin/users is 404. Teacher API exists with no web route.
10	Student transport profile	Route and homeStopId fields exist.	Only 1 of 3 students has a pickup stop set.
14	Route stop management	Stops can be created under a route, with order and map point.	No Transport → Stops menu and no /admin/stops list.
17	Bus-to-route assignment	Bus is assigned on the schedule / trip instance.	No standing bus-to-route assignment record.
18	Pickup / drop-off configuration	One home stop per student.	No separate pickup vs drop-off stop.
19	Transport dashboard	School dashboard has trip charts and counts.	Not a dedicated transport operations dashboard.

STEP	ITEM	WHAT WORKS	WHAT IS STILL INCOMPLETE
28	Student attendance / trip history	TripEvent records store check-in/out against the Trip ID.	No dedicated attendance report or per-student status page.
30	Transport notifications	Parent and driver in-app + push notifications work.	No Admin Notifications page. Header badge shows a fake count of 12.
31	Transport alerts	Driver approach/arrive toasts and late-pickup alerts.	No admin operational-alerts module.
32	Admin transport controls	Buses, drivers, routes, schedules, instances, cancel/override work.	Stops, campuses, and users are not as specified.
33	Student transport status	Live Tracking shows check-in/out counts.	No student-by-student status screen for the active trip.
34	Trip tracking & history	Instances, events, and GPS are retained.	No dedicated history/report view beyond the instances table.
35	Driver trip history	/driver/trips/completed returned 15 trips.	Driver App has no history list UI.
37	Transport data & API services	Core APIs for students, parents, buses, drivers, routes, trips, GPS, attendance work.	Missing campuses, classes, reports, audit, users, standalone stops APIs.
38	Database & relationships	School, User, Kid, Bus, Route, Stop, Schedule, Trip, Event, GPS, Holiday, Exception exist.	No Campus, Class, AcademicYear, or AuditLog entities.
42	Operational search & filtering	Trip Instances filter by date, status, and route.	Topbar "Search anything" is unused local state.
43	Production notifications & error handling	API validation and UI toasts exist.	Fake admin notification/message counts. CORS throws on unknown origins.
44	Testing & integration	This audit live-tested the scheduled Trip ID flow.	smoke-test.js still starts an ad-hoc trip, not a scheduled instance.
48	Route, stop, bus & driver management	Create/manage routes, buses, and drivers works.	Stops are nested inside Routes; no dedicated Stops page.
50	Trip scheduling & instances	ONE_TIME, EVERY_DAY, WEEKDAYS, CUSTOM_DAYS; morning/afternoon/evening; exceptions; holidays; edit scopes; unique Trip IDs. Same-period overlap blocked (tested).	Conflict is by whole period, not time window, so a bus cannot run two morning trips at different times. No duplicate. No scheduling audit history. Data currently only has morning trips.
51	Driver App trip execution	Today's Trip Instances, Start {period} trip, GPS, check-in/out, assigned-driver gate, open-check-in block — all tested.	Ad-hoc Start morning/evening still bypasses Trip IDs. Admin cannot operate a trip (403). No completed-history UI. Only one active trip per driver.

3. What is missing (7 steps)

These have no working module. APIs return 404.

STEP	ITEM	CURRENT STATE
5	Campus management	No Campus model, API, or UI. /admin/campuses is 404. School is a single location.
6	Academic year / term management	No AcademicYear or Term entity. /admin/academic-years is 404.
7	Class / grade management	Classes nav item is Coming Soon. Grade is free text on the student (Grade 4, Grade 3, 1).
40	Audit & activity logging	No AuditLog model, API, or UI. /admin/audit-logs is 404.
41	Admin reporting foundation	No reports routes or pages. Only leave-request CSV exists.
47	Student, class & campus assignment	No class or campus entities to assign. Grade is a string field only.
52	Production reports & analytics	/admin/reports and /admin/reports/analytics are 404. No executive dashboard, trip instance report, attendance report, performance reports, or trip CSV.

4. What still needs to be done

Build list, ordered by how blocking each item is against the handover document. Start at the top.

1. **Step 52 — Reports & Analytics.** Executive dashboard (scheduled, completed, in-progress, cancelled, pending, average duration, start delay, attendance). Trip Instance report. Student attendance per Trip ID with timestamps/locations. Driver, bus, and route performance. Recurring schedule performance and exceptions. Audit history for scheduling. CSV exports. Read-only — must not modify operational records.
2. **Admin navigation (spec section 4).** Required order: Dashboard → Students → Parents → Classes → Campuses → Transport → Live Tracking → Reports & Analytics → Notifications → Users & Roles → Settings → Audit Logs. Transport submenu: Buses, Drivers, Routes, Stops, Trip Scheduling, Trip Instances. Live Tracking must be a top-level item, not nested under Transport.
3. **Steps 5 + 47 — Campuses.** Create/manage campuses and assign students, routes, and trip instances to the correct campus.
4. **Steps 7 + 47 — Classes.** Class/grade entities (not free-text grade) and student-to-class assignment for scheduling and reporting.
5. **Step 6 — Academic year / term.** Configure academic years, terms, and operating periods.
6. **Step 40 — Audit logs.** Record important administrative and operational changes, including scheduling edits, with an Admin Audit Logs screen.
7. **Step 3 — Users & Roles.** Admin screen to create users, assign roles, and manage permissions. Add a teacher web route or drop the unused teacher API from the product surface.

8. **Step 30 — Admin Notifications.** Real notifications page for transport events. Remove fake header badges (12 notifications / 4 messages).
9. **Step 14 — Dedicated Stops page.** Transport → Stops: create, order, assign to routes.
10. **Step 50 — Time-window conflicts.** Allow a bus to run multiple morning (or afternoon/evening) trips when times do not overlap. Today the conflict check blocks any second trip in the same period.
11. **Step 50 — Duplicate and schedule audit.** Duplicate a recurring schedule. Keep an audit history of schedule changes. Historical completed trips must continue not to be silently rewritten.
12. **Step 18 — Separate pickup and drop-off stops.** Configure each student's pickup stop and drop-off stop independently. Also set home stops for the 2 of 3 students currently missing one.
13. **Steps 35 + 51 — Driver trip history UI.** Show assigned and completed trips in the Driver App (API already returns completed trips).
14. **Step 51 — Driver App alignment.** Remove ad-hoc Start morning / Start evening that creates trips without a scheduled Trip ID. Allow authorized admin roles to operate a trip. Keep Start/End Morning, Afternoon, and Evening controls tied to the correct Trip ID.
15. **Step 42 — Operational search.** Wire the topbar search so it actually filters students, parents, routes, buses, drivers, and trips.
16. **Step 2 — Password change / reset.** Self-service password change. Admins can already set a password on create/edit.
17. **Step 41 / 28 / 33 / 34 — Reporting data surfaces.** Student attendance per Trip ID, trip history report, and student transport status for the applicable trip (feeds Step 52).
18. **Step 44 — Update smoke tests.** Replace ad-hoc trip start with the scheduled Trip ID flow used in production.

5. Live tests run on 17 August 2026

CHECK	RESULT	DETAIL
API health	PASS	Local API on :4001 and Render production both returned ok
Login — 5 roles	PASS	school_admin, driver, parent, super_admin, teacher
Invalid password	PASS	401
RBAC walls	PASS	Driver/parent cannot hit admin; admin cannot hit driver/parent APIs
CRUD modules	PASS	Dashboard, school, students, parents, drivers, buses, routes, schedules, instances, holidays, leave, noticeboard
Same-period conflict	PASS	Second morning trip for KDA 123A / Daniel created 0 instances, 1 conflict
Start TRIP-1005	PASS	Status active; start GPS stored
Live map while active	PASS	KDA 123A, Daniel Kamau, morning, to school, Route A; in=0 then in=1
GPS ping	PASS	lat/lng/heading/speed persisted

CHECK	RESULT	DETAIL
Check-in / check-out	PASS	picked_up then dropped_off on that Trip ID
Block end with open check-in	PASS	409 Cannot complete trip while students are still checked in
End trip + leave live map	PASS	Completed; live buses = 0; restart rejected with 400
Parent notifications	PASS	trip_started, kid_picked_up, kid_dropped_off, trip_completed
Campuses / classes / reports / audit / users	MISSING	All 404
Admin start trip	GAP	403 — spec allows authorized admin roles; code is driver-only

6. Admin navigation — required vs current

Required

#	MENU ITEM
1	Dashboard
2	Students
3	Parents
4	Classes
5	Campuses
6	Transport
7	Live Tracking (top-level)
8	Reports & Analytics
9	Notifications
10	Users & Roles
11	Settings
12	Audit Logs

Transport submenu: Buses · Drivers · Routes · Stops · Trip Scheduling · Trip Instances

Current

#	MENU ITEM
1	Dashboard
2	School
3	Students
4	Parents
5	Staff
6	Classes (coming soon)
7	Subjects / Examinations / Assignments (coming soon)
8	Attendance / Leave / Bulk attendance
9	Noticeboard / Messages
10	Transport (includes Live Tracking)

Transport submenu: Routes · Drivers · Vehicles · Trip Scheduling · Trip Instances · Live Tracking (should be top-level)

Live Tracking is in the wrong place. The spec requires a dedicated top-level Admin menu item, separate from Transport scheduling and Reports & Analytics.

7. Recommended first implementation slice

Start with **Reports & Analytics (Step 52)** and **moving Live Tracking to a top-level menu**. Those are specified as production surfaces: one is entirely absent, the other is misplaced.

SchoolKids Tracker — handover status report · 17 August 2026 · 52 steps: 24 done, 21 partial, 7 missing.